



Guide for the developers of the iEPB schema

WP 3, T 3.1

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1. Executive summary

Within the framework of the European iEPB project, which aims to facilitate the effective implementation of energy performance assessments of buildings included in the EPBD, the iEPB schema is defined. This schema is a common data format that aims to harmonize the different energy assessment tools, such as Energy Performance Certificates (EPC), the Smart Readiness Indicator (SRI), the Building Renovation Passport (BRP), and the Life Cycle Global Warming Potential (LCGWP), promoting consistent implementation of the EPBD in Member States. Deliverable D3.1 defines the guidelines governing the development of this iEPB schema.

Among the main objectives of this document are: to define a common file format that allows multiple EPB assessments to be stored in a single container and information to be exchanged between them; to establish common criteria for data definition, ensuring consistency between countries and tools; laying the foundations for facilitating the import and export of data between the iEPB format and EPB assessment tools (being the national EPC demonstration software to be implemented in the iEPB schema CE3X, CERMA, VABI and ECOTECH); and enabling the scalability and extension of the schema to include future assessments and tools.

The iEPB schema is extensible and scalable, making it a living, open schema for participation and collaboration with any other EPB assessment tool or other building assessments. Therefore, at the end of the project, the iEPB schema will be included in the public repository [iEPBproject/iEPBschema \(github.com\)](https://github.com/iEPBproject/iEPBschema), where it will be accessible to developers of other assessment tools who wish to join the format.

The iEPB schema is structured as a compressed file with the extension .iEPB, containing four main components: a gbXML.xml file with the building's energy model, an iEPB.xml file with supplementary information, an ILCD folder with data for assessing the climate impact of the life cycle, and an optional "Source" folder for additional images and documents. This structure allows multiple assessments to be stored in a single file.

From a technical perspective, the schema is based on the XML language, following standardized syntax rules and clear naming conventions. Precise rules are set to decide when to use elements or attributes, based on structure, readability, and processing criteria. Plus, specific blocks are defined for each country, assessment, and assessment tool, making it possible to include multiple assessments for the same building.

This deliverable is supported by the results of document *D2.1 – Transparent Description of Each National and EU Technical Methodology for EPB Assessment*, which provides a detailed analysis of the existing regulatory and methodological frameworks in Austria, the Netherlands and Spain. This analysis served as the basis for identifying the differences and convergences between national methodologies and for defining the functional requirements of the iEPB schema, ensuring its applicability both in environments with centralised and highly integrated databases and in those where energy registers are fragmented or managed in a decentralised manner.

The document includes examples of the software CE3X (Spain), Ecotech (Austria) and the SRI2Market web tool implemented in the iEPB schema. These examples show how the different types of data included in these tools are integrated into the iEPB format (administrative, envelope, installations, etc.) and specify which information should be stored in the gbXML.xml file and which in the iEPB.xml file in the shared blocks or in those specific to each tool, ensuring the correct interoperability of data between the different assessments. The examples also describe some of the challenges identified and how they have been addressed.

Finally, the iEPB schema is presented as a scalable and open solution that can be adopted by new tools and countries in the future. Its implementation will contribute to improving interoperability between building energy assessment software (EPC, SRI, BRP, etc.), reducing errors in data management, facilitating the reuse of information, and promoting collaboration between software developers.



2. iEPB schema

The iEPB schema is the common data file proposed in the iEPB project to achieve the project's final objective of promoting the effective integration of EPB assessments, namely between existing Energy Performance Certificates (EPCs) and other complementary instruments, in particular the Smart Readiness Indicator (SRI) and the energy renovation recommendations.

The data model developed (iEPB schema) aims to be common for the EU in order to increase consistency and harmonisation between Member States. Furthermore, it aims to be expanded and scaled up in the future to allow for the integration of additional EPB assessments.

The implementation of this common data file in different EPC tools by different entities, as well as their desire for scalability, requires the establishment of a common set of rules for the definition of the data required for each of the assessments that wish to subscribe to the format. This document has been developed in order to establish the common criteria to be followed by any EPB assessment tool to implement its data in the iEPB schema. It is the document being followed by the developers of the CE3X, CERMA, VABI and ECOTECH tools to implement their tools in the iEPB schema.

It is good practice to coordinate the implementation of the data in the schema together with the implementation of its import and export to the assessment tool itself. This coordination avoids errors and optimises efforts by avoiding duplication of work. For this reason, the current iEPB schema is in the process of definition and its final version will be the one obtained at the end of Task 4.2. Its scope includes ensuring that the national energy certification software (CE3X, CERMA, VABI and ECOTECH) can import the necessary data from iEPB schema to carry out an official EPB assessment and that the iEPB schema is able to integrate the output data from these assessments.

On the other hand, the fact that the iEPB schema is extendable and scalable makes it a living and open schema for involvement and collaboration with any other EPB assessment tool. To make this possible, the iEPB schema has been included in the [iEPBproject/iEPBschema \(github.com\)](https://github.com/iEPBproject/iEPBschema) repository. This repository, currently private, will become public once the above certification softwares are implemented and will be managed by CENER and EFINOVATIC. From here, it will be accessible to developers of other assessment tools whose data, in turn, can be used.



3. Definitions and general structure of the iEPB schema

3.1. iEPB data format

The iEPB format is a multiformat, primarily based on the current gbXML file format, which in turn is an open-source file format for the exchange of building information based on the World Wide Web Consortium's XML standard.

This format will be able to store the different EPBD assessments in just one file:

- Energy Performance Certificate (EPC)
- Smart Readiness Indicator (SRI)
- Life Cycle Global Warming Potential (LCGWP)
- Building Renovation Passport (BRP)

For each of the assessments, the format will have the possibility to store the current state, the previous states of past evaluations and the possible renovated states if improvement measures are analyzed.

3.1.1. General structure of the iEPB data format

The iEPB schema will consist on three documents:

1. **gbXML file:** File containing the Building Energy Model (BEM) in the Green Building XML Schema format.
2. **iEPBxml file:** This file will host all the information that cannot be stored in the previous file.
3. **ILCD folder:** Folder containing the required data for the Life Cycle Global Warming Potential assessments in the International Reference Life Cycle Data System (ILCD format).

The iEPB schema could also include an auxiliary folder:

4. **“Source”** optional folder where images and other documents required by the assessments can be stored.
Images, documents and other binary files can be stored in this folder or embed in an element iEPBxml using Base64 encoding standard.

These two files and the two folders will be stored into one file using zip technology according to the ISO/IEC 21320-1:2015. The zip file extension will be renamed to **.iEPB**.

3.1.2. Characteristics of the iEPB file

The iEPB file must comply with the following characteristics to be compatible among software and show features of similarity regardless of the software it comes from.

3.1.2.1. Naming convention

The files included in the iEPB format shall be named according to the rules set out in the following table:

Table 1: Naming rules for the iEPB format.

File/folder	Name
iEPB	NameXX.iEPB
gbXML file	NameXX_gbXML.xml
ILCD folder	ILCD
iEPBxml file	NameXX_iEPB.xml
folder	Source

In that way and as an example, if the iEPB file is called: “**OfficeBuilding1.iEPB**” when this file is decompressed, the files and folders contained in it will be named: “OfficeBuilding1_gbXML.xml”, “OfficeBuilding1_iEPB.xml”, “ILCD” folder and finally an optional folder named “Source”.

NOTE: Although the code for naming the files by the software is as above, the import program should not use the iEPB file name to locate the gbXML file and the iEPB file. To do this, look for files ending in “_gbXML.xml”, “_iEPB.xml”. Otherwise, a renamed iEPB file cannot be used.

3.1.2.2. iEPB file zip compression algorithm

In the case of the iEPB file, the functionality of the zip format is not to reduce the size of the files that compose it, but to merge them into a single file. For this reason, the compression algorithm to be used for the iEPB format will be: “**Store without compression**”

It is true that the iEPB file could become a very large file if ALL the building data were stored in it, however, at present its size is moderate and since there is currently no standalone tool with which to read the iEPB files, the ‘Store without compression’ algorithm allows the data contained in the iEPB file to be viewed with a tool that reads the xml format.

3.1.3. New iEPB file

The concept behind the iEPB format is to incorporate all information relating to the building's EPB assessments (or others) into a single file. The initial idea was to be able to store several assessments of the same type in the same iEPB file (e.g. EPC assessment before renovation and EPC assessment after renovation), but after various discussions on the different scenarios that could arise, it was concluded that each file could only store one assessment of each type. Managing a new assessment through the iEPB web app. In the platform, a ‘Building’ can contain several “Versions” and be managed. Each version is an iEPB file. Therefore, after generating a new ‘Version’ for the building, the changes made can be recorded in this new version.

This solves problems such as, for example, the renovation of a building in which its envelope is modified by increasing the percentage of openings: a new version of the building will define a new iEPB file and therefore a new gbXML (shared for all assessments included in the same version) that will not affect data from previous assessments that are fed from it. The version-based solution allows an assessor to make a copy of the desired data from a previous assessment (e.g. an EPC), edit it and/or add and/or delete data in the new version without compromising the authorship and results of the initial assessment.

3.1.4. XML syntax: general rules

The general XML rules apply to the documents included in the iEPB format: gbXML and iEPBxml. The Extensible Markup Language (XML) is a text-based file that stores data in the form of hierarchical elements. The XML specifications are defined by the W3C recommendation: Extensible Markup Language (XML) 1.0 (Fifth edition) that can be found in (<https://www.w3.org/TR/2008/REC-xml-20081126/>). One of the most important characteristics of this format is that it is both human-readable and machine-readable.

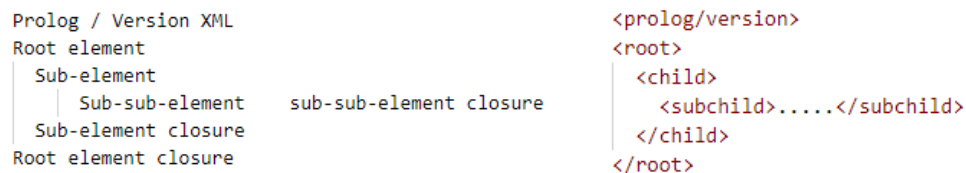


Figure 1: Basic structure of an xml

The syntax rules of XML are very simple and logical, easy to learn and use. The XML document is composed by **XML Elements**. An XML element is everything from (including) the element's start tag to (including) the element's end tag. XML documents must contain one **root element** that is the parent of all other elements. An element can contain: text, attributes, other elements or a mix of the above.

Usually, **countries, buildings, assessments** and **tools**, according to the xml syntax, **are elements**, **however**, in this document, in order to differentiate these elements of a more principal nature than the rest, when we refer to them **we will refer to them by the word "Block"**. The rest of the elements contained in these "blocks" will be referred to as elements as indicated in the official nomenclature.

3.2. Definition of the iEPB.xml file

3.2.1. Common criteria to define the iEPB.xml

To give homogeneity and harmonization to all data collected in the different blocks (country blocks) of the iEPB schema, the following criteria will be used for the definition of the iEPBxml:

- Version 1.0 of the xml will be used.
- The encoding to be used is UTF-8.
- The elements of the country-specific blocks will be in English.
- The description of the elements (in the desired language) will be in the xsd (in the definition file). This description is what the portal users will see when they enter the data. For example:

```
<xs:annotation>
  <xs:appinfo>CE3X</xs:appinfo>
  <xs:documentation xml:lang="es">
    Referencia catastral
  </xs:documentation>
  <xs:documentation xml:lang="ca">
    Referència cadastral
  </xs:documentation>
</xs:annotation>
```

Each variable will be defined by attributes referring to the language of its country block.

- Element naming convention will follow **PascalCase** style (words are limited by capital letters, e.g. VariableOne, VariableTwo...) for elements but in case of acronyms will be in capital letters (for example EPC (Energy Performance Certificate) or SRI , HVAC....)
- The name of the attributes shall follow the **camelCase** style.
- The images and other documents to be included in the tools will be called in a folder called *external docs*
- Elements for specific software, if possible, will use the convention adopted for naming, but it will be possible to defined like the name of the software respecting capital letters, for example CE3X

3.2.2. Usage of elements or attributes in the iEPBxml file

This section defines the rules to be considered when defining a data as an element or attribute. Sometimes the information can be stored in both an element or an attribute. Here are some recommendations:

- If the information can subdivide in more elements up with elements, use element.
- If the information is suitable for attribute form, but could end up as multiple attributes of the same name on the same element, use child elements instead.
- If the information is required to be in a standard DTD-like attribute type such as ID, IDREF, or ENTITY, use an attribute.
- If the information should not be normalized for white space, use elements. (XML processors normalize attributes in ways that can change the raw text of the attribute value.)

DTD principle (document type definition)

- Use elements with CDATA
- Use attributes when you need ID, IDREF or ENTITY.

Basic content principle



- If you consider the data to be an essential part of the XML, use element.
- If you consider the data to be secondary, or only useful to help applications process the main communication, use attribute.

Questionable example	<pre><person> <id>1</id> <Name>Marta</Name> <FamilyName>Sampedro</ FamilyName > </person></pre>
Example of how to do it in the iEPB schema	<pre><person id="1"> <Name>Marta</Name> <FamilyName>Sampedro</ FamilyName > </person></pre>

Structured information principle

- If you want the information to be expressed in a structured form, especially if the structure may be extensible, use elements.
- If you want the information to be expressed in an atomic token, use attributes.

Example with structured info	<pre><person id="1"> <Name>Marta</Name> <FamilyName>Sampedro</ FamilyName > <Address> <Street>Carlos III</Street> <Number>7</Number> <City>Pamplona</City> </Address> </person></pre>
Example with abbreviated info	<pre><person id="1"> <Name>Marta</Name> <FamilyName>Sampedro</ FamilyName > <Address street="Carlos III" number="7" city="Pamplona"></Address> </person></pre>

Readability principle

- If the information is to be processed by a person, use elements.
- If the information is to be digested by a machine, use attributes.

Example of information for people	<pre><person id="1"> <Name>Marta</Name> <FamilyName>Sampedro</ FamilyName > <Address> <Street>Carlos III</Street> <Number>7</Number> <City>Pamplona</City> </Address> </person></pre>
Example with information for programmes	<pre><person id="1" name="Marta" familyName="Sampedro" street="Carlos III" number="7" city="Pamplona"></person></pre>

Element-attribute binding principle

- Use elements if you need its value to be modified by another attribute.
- Never use attributes that modify other attributes.

Example where one element modifies another element	<pre> <person> <Balance>200</Balance> <CoinType>EUR</CoinType> <Name>Miguel</Name> <FamilyName>Suárez</FamilyName> </person> </pre>
Example where one attribute modifies another attribute	<pre> <person balance="200" coinType="EUR"> < Name > Miguel </ Name > < FamilyName > Suárez </ FamilyName > </person> </pre>
Example where one attribute modifies another element	<pre> <person> <Balance coinType="EUR">200</Balance> < Name > Miguel </ Name > <FamilyName > Suárez </ FamilyName > </person> </pre>

In the iEPB schema attributes will be used to define:

- **id**
- **unit**
- **lang** (language)
- **constructionType**

In each tool shall be included:

- **order** (in case there are more than one assessment of the same type)
- **vigent**="True"

3.2.3. Common content of iEPBxml file

The root element of the iEPBxml file will be the iEPBxml. This root element will have a **child for each** of the three **countries** that participate in the iEPB project and, subsequently, there may be one for each country that wishes to join the format later.

```

<iEPBxml>
  <Spain>
  </Spain>
</iEPBxml>

```

Each of the child country elements shall be composed of the following subchilds:

- building block containing common immutable information related to the building, that is common to the different assessments.
- a subchild for each of the possible assessment types (EPC, SRI, LCGWP, ...)



```

<iEPBxml>
  <Spain>
    <Building></Building>
    <EPC></EPC>
    <SRI></SRI>
    <BRP></BRP>
  </Spain>
</iEPBxml>

```

Each assessment type (e.g., EPC) will contain the following subchilds:

- an element for each assessment tool used to perform the assessment. For example, in Spain, for the EPC assessment, we could have used two different tools: CE3X or CERMA. The information required by CE3X will be stored in the CE3X and the information required by CERMA will be stored in the CERMA block. If another tool wants to join the iEPB schema each tool will have its own specific block.
- an evaluation results element called <Results> that is common to all software included in the evaluation and will be defined by national requirements

```

<iEPBxml>
  <Spain>
    <Building></Building>
    <EPC>
      <CE3X></CE3X>
      <CERMA></CERMA>
      <Results></ Results>
    </EPC>
    <SRI></SRI>
    <DRP></DRP>
  </Spain>
  <Austria>
    <Building></Building>
    <EPC>
      <ECOTECH></ECOTECH>
      <Results></ Results>
    </EPC>
    <SRI></SRI>
    <DRP></DRP>
  </Austria>
  <Netherlands>
    <Building></Building>
    <EPC>
      <VABI></VABI>
      <Results></ Results>
    </EPC>
    <SRI></SRI>
    <DRP></DRP>
  </Netherlands>
</iEPBxml>

```

Figure 2: Basic structure for the iEPBxml file

Finally, within each block of tools::

- Each assessment entry will contain the complete set of information needed to perform the calculation. (Note that the assessment results will be contained at the top level as they are common to all assessments of the same type).
- Information related to the BEM will be preferably stored in the already defined elements of gbXML.xml file. If the existing elements in the gbXML are not appropriate for this information, a new element will be created in the iEPB.xml file.

Multiple building assessments can be stored in the same iEPB file, provided that the assessments share all the data stored in shared blocks or files such as “Building,” “Results,” or the gbXML.xml file. Multiple assessments of the same type should not be included in the same iEPB file, as the results block is common and shared.

When more than one assessment of the same type is available for a building (e.g. before and after renovation), the assessments will be managed through the iEPB web app by generating different versions (each version corresponds to an iEPB file) within the same building . The creation of new versions facilitates the understanding of the data stored in the iEPB file, preventing the duplication of fields in the same file and guaranteeing the authorship of the technical assessors.

```
<iEPBxml>
  <Spain>
    <Building id="Building1"></Building>
      <CadastralReference>310000000001869003HK</CadastralReference>
      <City>Pamplona</City>
      <State>Navarra</State>
      <ApplicableLegislation>NBE-CT-79</ApplicableLegislation>
      <YearOfConstruction>1976</YearOfConstruction>
      <Tipology>Unifamiliar</Tipology>
      <UsageProfile>Residential</UsageProfile>
      <HE1ClimaticZone>D1</HE1ClimaticZone>
      <HE4ClimaticZone>II</HE4ClimaticZone>
      <NominalFloorToCeilingHeight>2.5</NominalFloorToCeilingHeight>
      <NumberOfStories>2</NumberOfStories>
      <AirChangesPerHour>0.45</AirChangesPerHour>
      <DHWDailyDemand>126</DHWDailyDemand>
      <InternalPartitionMass>Media</InternalPartitionMass>
      <LocationImage>ubicacion.jpg</LocationImage>
      <BuildingImage>edificio.jpg</BuildingImage>
      <AirtightnessTest>False</AirtightnessTest>
      <BlowerDoorTestQ50></BlowerDoorTestQ50>
      <BlowerDoorTestN></BlowerDoorTestN>
    </Building>
    <EPC>
      <CE3X>
        <Building id="Building1"></Building>
          <NominalFloorToCeilingHeight>2.4</NominalFloorToCeilingHeight>
          <AirChangesPerHour>0.38</AirChangesPerHour>
        </CE3X>
      <CERMA></CERMA>
      <Results></Results>
    </EPC>
    <SRI></SRI>
    <DRP></DRP>
  </Spain>
</iEPBxml>
```

Figure 3: Example of two EPCs of the same building changing the value of <NominalFloorToCeilingHeight> and <AirChangesPerHour>

3.3. Definition of the gbXML.xml file

The Green Building XML schema (gbXML) is an open schema developed to facilitate transfer of building data stored in Building Information Models (BIM) to engineering analysis tools. gbXML is streamlined to transfer building properties to and from engineering analysis tools to reduce the interoperability issues and eliminate plan take-off time.

The gbXML schema is a very advanced schema for the geometrical definition of the building so, as far as possible, the building geometry will be defined in the gbXML data format.

In the case of air-conditioning systems, lighting, etc., the elements defined in gbXML shall be used and, if this is not possible, they shall be defined in the blocks corresponding to the tool and country to which they belong.

In summary, all building data (geometry, building systems, HVAC systems, operating schedules...) that can be defined in the common gbXML data file will be defined in this file.

All the information needed to define the import and export data of the tools to the gbXML file can be found in the public github repositories of the Green Building XML (gbXML) [Green Building XML \(gbXML\) · GitHub](#) . Version gbXML 7.03 will be used ([GreenBuildingXML Ver7.03 \(gbxml.org\)](#))

3.4. Definition of the ILCD folder

The global warming potential (GWP) over a building's whole life cycle indicates the building's overall contribution to emissions that lead to climate change. It brings together greenhouse gas emissions embodied in construction products with direct and indirect emissions from the use stage. A requirement to calculate the life-cycle GWP of new buildings therefore constitutes a first step towards increased consideration of the whole life-cycle performance of buildings and a circular economy.

The new EPBD recast approved last 24 April 2024 indicates that the energy performance certificate for buildings must provide information on their primary and final energy consumption, energy needs, renewable energy production, greenhouse gas emissions and life cycle GWP if available. Since the iEPB schema is a common data file for EPB assessments, it has been considered necessary to include the data related to life cycle GWP in the *.iEPB format and for this purpose, an ILCD folder is included in it.

The International Reference Life Cycle Data System (ILCD) is an initiative developed by the JRC and DG ENV, with the objective of providing guidance and standards for greater consistency and quality assurance in the application of Life Cycle Analysis (LCA).

In order to build on the path already taken, the iEPB schema will integrate in its composition an ILCD folder elaborated following the format specifications included in the European Commission website: <https://eplca.jrc.ec.europa.eu/LCDN/developerILCDDataFormat.html>.

The ILCD folder will include the information related with the Environmental Product Declarations involved: contacts, external documents, flow properties, flows, processes, sources and unit groups. This folder will also include the information regarding the life cycle model (eILCD).

4. Examples

This section includes examples of the implementation of the energy certification software CE3X (official tool in Spain) and ECOTECH (official tool in Austria) and the SRI assessment tool SRI2MARKET.

4.1. Spanish software CE3X to iEPB

The official Spanish energy certification software CE3X was the first software to be implemented into the iEPB schema and served as a guide for the implementation of other software. The following is a detailed example of how it was implemented.

4.1.1. Administration data

The CE3X tool includes three sections regarding administration data. These three sections are:

- Location and building identification.
- Customer data.
- Details of the certifying technician.

CE3X - RES: Certificación energética simplificada de edificios existentes - Residencial

Archivo Librerías Patrones de sombra Resultados Complementos Ayuda Acerca de

Datos administrativos Datos generales Envolverte térmica Instalaciones

Localización e identificación del edificio

Nombre del edificio

Dirección

Provincia/Ciudad autónoma Localidad Código Postal

Referencia Catastral +

Datos del cliente

Nombre o razón social

Dirección

Provincia/Ciudad autónoma Localidad Código Postal

Teléfono E-mail

Datos del técnico certificador

Nombre y Apellidos NIF

Razón social CIF

Dirección

Provincia/Ciudad autónoma Localidad Código Postal

Teléfono E-mail

Titulación habilitante según normativa vigente

Figure 4: CE3X Tool Administration data tab

The way to introduce each of this sections in the iEPB format is as follows:

4.1.1.1. Location and building identification

Localización e identificación del edificio

Nombre del edificio	Edificio Singular		
Dirección	Avda. Ejército		
Provincia/Ciudad autónoma	Navarra	Localidad	Pamplona
Código Postal	31001		
Referencia Catastral	310000000001869003HK	+	

Figure 5: Location and building identification data introduced in CE3X tool

These data refer to the data of the building model to be assessed. In gbXML they correspond to the element <Campus>. Within an assessment or <Campus> there may be one or more buildings which are referred to as <Building>.

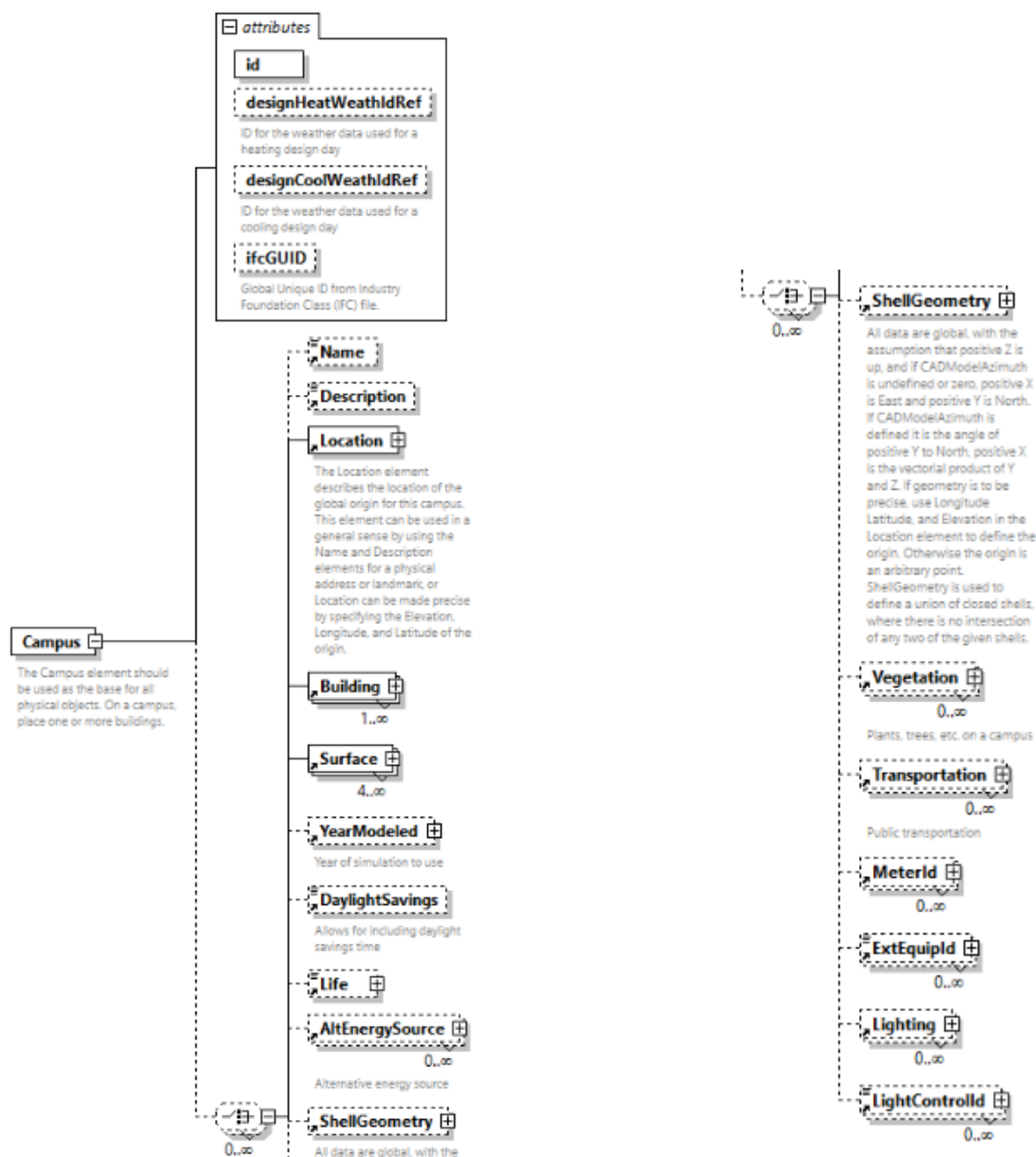


Figure 6: Diagram of the <Campus> element of the gbXML schema

```

8  <Campus id="Facility">
9    <Name>Edificio Singular</Name>
10  <Location>
11    <ZipcodeOrPostalCode>31001</ZipcodeOrPostalCode>
12  </Location>
13  <Building id="Building1" buildingType="MultiFamily">
14    <Name>Building 1</Name>
15    <StreetAddress>Avda. Ejército</StreetAddress>
16    <Area>4815.37</Area>
17  <Space id="Space_1" zoneIdRef="TZ__Space_1">...
23  </Space>
24 </Building>

```

Figure 7: <Campus> element defined in the gbXML schema used for defining Location and building identification data of the CE3X tool

The <Name> of the Campus is the CE3X *Nombre del edificio*, the <ZipcodeOrPostalCode> is the *Código Postal*, and within <Building> <StreetAddress> shall be defined for *Dirección* and also the buildingType included as an attribute of the <Building> element.

The rest of the fields (*Provincia (State)*, *Localidad (City)* and *Referencia catastral (Cadastral Reference)*) included in the CE3X tool and not present in the gbXML will be determined in the "BuildingData" block created in the Spanish section as they are not specific data of any tool but characteristics of the building itself.

```

<Spain>
  <Building id="Building1">
    <CadastralReference>310000000001869003HK</CadastralReference>
    <City>Pamplona</City>
    <State>Navarra</State>
    <ApplicableLegislation>NBE-CT-79</ApplicableLegislation>
    <ConstructionYear>1976</ConstructionYear>
    <Tipology>Unifamiliar</Tipology>
    <UsageProfile></UsageProfile>
    <HE1ClimaticZone>D1</HE1ClimaticZone>
    <HE4ClimaticZone>II</HE4ClimaticZone>
    <NominalFloorToCeilingHeight>2.5</NominalFloorToCeilingHeight>
    <NumberOfStories>2</NumberOfStories>
    <AirChangesPerHour>0.45</AirChangesPerHour>
    <DHWDailyDemand>126</DHWDailyDemand>
    <InternalPartitionMass>Media</InternalPartitionMass>
    <LocationImage>ubicacion.jpg</LocationImage>
    <BuildingImage>edificio.jpg</BuildingImage>
    <AirtightnessTest>False</AirtightnessTest>
    <BlowerDoorTestQ50></BlowerDoorTestQ50>
    <BlowerDoorTestN></BlowerDoorTestN>
  </Building>
  <SRI></SRI>
</EPC>

```

Figure 8: <BuildingData> element defined in the Spanish block

In this <Building> block, more data related to the building characteristics will be included as can be seen in the Figure 8.

4.1.1.2. Customer data

Datos del cliente

Nombre o razón social	Javier González		
Dirección	Plaza del Castillo		
Provincia/Ciudad autónoma	Navarra	Localidad	Pamplona
		Código Postal	31001
Teléfono	666123456	E-mail	cliente@ce3x.com

Figure 9: Customer data introduced in CE3X tool

Customer data are only required in CE3X tool. The gbXML schema does not contain any element that collects customer data and it is not necessary to include this type of data in other Spanish assessments (not even in other EPC assessments). Moreover, the client contracting a CE3X assessment does not necessarily have to be the same client contracting other assessments. Therefore this field will be within the specific block of the CE3X tool.

```

8      <ce3x>
9      |
10     |   <DatosCliente>
11     |   |   <Nombre>Javier González</Nombre>
12     |   |   <Direccion>Plaza del Castillo</Direccion>
13     |   |   <Provincia>Navarra</Provincia>
14     |   |   <Localidad>Pamplona</Localidad>
15     |   |   <CodigoPostal>31001</CodigoPostal>
16     |   |   <Telefono>666123456</Telefono>
17     |   |   <e-mail>cliente@ce3x.com</e-mail>
18     |   </DatosCliente>
19     |   <Construction constructionType="FachadaExteriorEstimada">
20     |   |   <CE3X order = "0">
21     |   |   |
22     |   |   |   <CustomerData>
23     |   |   |   |   <Name>Javier González</Name>
24     |   |   |   |   <Address>Plaza del Castillo</Address>
25     |   |   |   |   <State>Navarra</State>
26     |   |   |   |   <City>Pamplona</City>
27     |   |   |   |   <ZipcodeOrPostalCode>31001</ZipcodeOrPostalCode>
28     |   |   |   |   <PhoneNumber>666123456</PhoneNumber>
29     |   |   |   |   <e-mail>cliente@ce3x.com</e-mail>
30     |   |   |   </CustomerData>
31     |   |   </CE3X>
32     |   </Construction>
33     </ce3x>

```

Figure 10: <DatosCliente> element defined in the Spanish block of the ce3x tool

4.1.1.3. Certifying technician data

Datos del técnico certificador

Nombre y Apellidos	Marta y Miguel Ángel	NIF	
Razón social	Cener - Efinovatic	CIF	B71107734
Dirección	C/ Melancolía		
Provincia/Ciudad autónoma	Navarra	Localidad	Pamplona
		Código Postal	31010
Teléfono	948316829	E-mail	info@efinovatic.es
Titulación habilitante según normativa vigente	Arquitecta		

Figure 11: Certifying technician data introduced in CE3X tool

This section contains the details of the qualified technician who is carrying out the energy performance certification of the building and who will be responsible for the data included in the certification. The gbXML schema includes an element called `<PersonInfo>` that contains information about people that created and modified the file. The technician who carried out the energy performance certificate of the building shall be one of these technicians, so this data shall be recorded in the gbXML block.

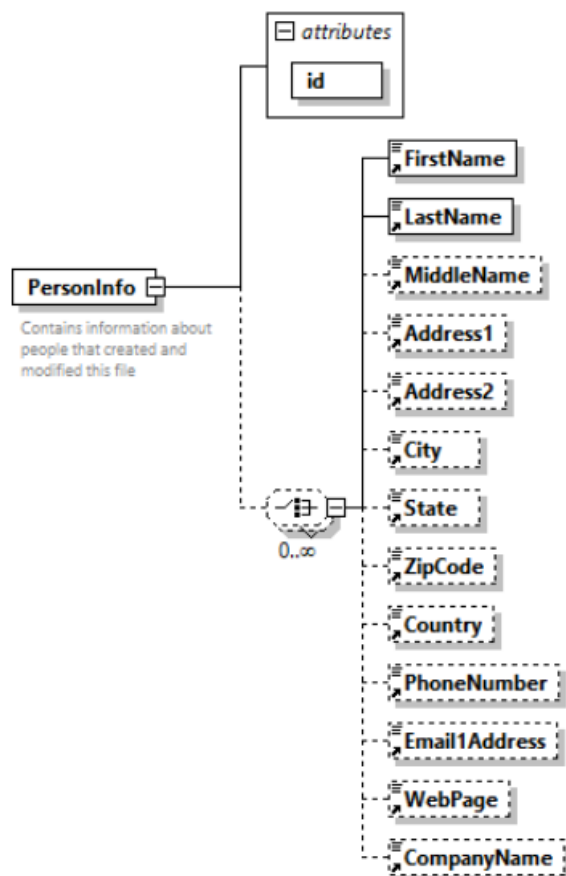


Figure 12: Diagram of the `<PersonInfo>` element of the gbXML schema

```

144 <PersonInfo id="certificadorEnergeticoID">
145   <FirstName>Marta y Miguel Ángel</FirstName>
146   <LastName>Unknown</LastName>
147   <Address1>C/ Melancolía</Address1>
148   <City>Pamplona</City>
149   <State>Navarra</State>
150   <ZipCode>31010</ZipCode>
151   <PhoneNumber>948316829</PhoneNumber>
152   <Email1Address>info@efinovatic.es</Email1Address>
153   <CompanyName>Cener - Efinovatic</CompanyName>
154 </PersonInfo>
  
```

Figure 13: `<PersonInfo>` element defined in the gbXML schema for the EPC

Many of the certifying technician's data have been defined in the gbXML but there are 3 data that cannot be defined which correspond to the VAT number (CIF), the Tax Identification Number (NIF) and the qualifying degree (titulación habilitante) of the certifying technician. These three blocks will be defined in the Spanish EPC block as they are required in all building energy performance certification tools. They will be defined within an element also called `<PersonInfo>` which will be

identified with the same id as the one contained in the gbXML and which will contain these three elements: <Cif>, <Nif> and <QualifyingDegree>.

```

1  <Spain>
2  ...
3  <epc>
4  <PersonInfo id="certificadorEnergeticoID">
5  <Cif>B71107734</Cif>
6  <Nif></Nif>
7  <TitulacionHabilitante>Arquitecta</TitulacionHabilitante>
  </PersonInfo>

```

Figure 14: <PersonInfo> element defined in the Spanish block of the EPC

<PersonInfo> element is used by <DocumentHistory> element of the gbXML. <DocumentHistory> need to be defined by at least two of the four elements it contains. Therefore, the elements PersonInfo, <CreatedBy> (which will record the creation of the file) and ProgramInfo will be defined.

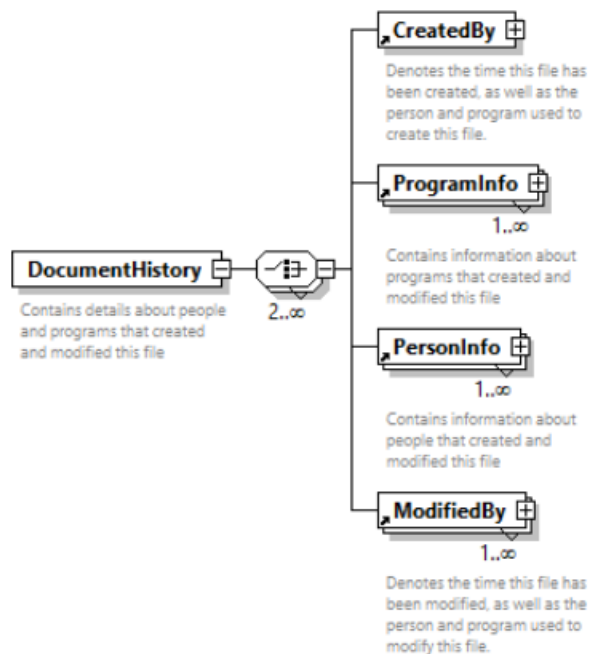


Figure 15: Diagram of the <DocumentHistory> element of the gbXML schema

```

140 <DocumentHistory>
141   <CreatedBy personId="certificadorEnergeticoID" programId="programaCertificacionID"
142     date="2024-03-25T09:41:00+02:00">
143   </CreatedBy>
144   <ProgramInfo id="ce3x">
145     <ProductName>CE3X</ProductName>
146     <Version>V2.3</Version>
147     <Platform>Windows</Platform>
148   </ProgramInfo>
149   <PersonInfo id="certificadorEnergeticoID">...
150 </PersonInfo>
151   <ProgramInfo id="openstudio">...
152 </ProgramInfo>
153   <PersonInfo id="openStudioID">
154   </PersonInfo>
155 </DocumentHistory>
156 </gbXML>

```

Figure 16: <DocumentHistory> element in the gbXML schema

4.1.2. General data

The General Data tab in CE3X is composed of two sections: *Datos generales* (general data) and *Definición edificio* (building definition). Both sections contain characteristics that are inherent to the building or its regulatory compliance and that will be shared with other assessments.

Datos administrativos Datos generales Envolverte térmica Instalaciones

Datos generales

Normativa vigente: NBE-CT-79 ? Año construcción: 1976

Tipo de edificio: Unifamiliar

Provincia/Ciudad autónoma: Navarra Localidad: Pamplona Zona climática: HE-1 D1 HE-4 II

Definición edificio

Superficie útil habitable: 173 m²

Altura libre de planta: 2.5 m

Número de plantas habitables: 2

Ventilación del inmueble: 0.45 ren/h

Demanda diaria de ACS: 126 l/día

Masa de las particiones internas: Media

☐ Se ha ensayado la estanqueidad del edificio

Imagen edificio Plano situación

Figure 17: General data tab of the CE3X tool

4.1.2.1. General data and building definition

"General Data" section includes building data such as: the type of building, year of construction, use and location. Most of these data will be allocated in the common element <Building> allocated in the Spanish block, and only the building type info will be allocated in the <Building> element allocated in the gbXML

```

<Spain>
  <Building id="Building1" buildingType="One flat">
    <CadastralReference>310000000001869003HK</CadastralReference>
    <City>Pamplona</City>
    <State>Navarra</State>
    <ApplicableLegislation>NBE-CT-79</ApplicableLegislation>
    <ConstructionYear>1976</ConstructionYear>
    <UsageProfile></UsageProfile>
    <HE1ClimaticZone>D1</HE1ClimaticZone>
    <HE4ClimaticZone>II</HE4ClimaticZone>
    <NominalFloorToCeilingHeight>2.5</NominalFloorToCeilingHeight>
    <NumberOfStoreis>2</NumberOfStoreis>
    <AirChangesPerHour>0.45</AirChangesPerHour>
    <DHWDailyDemand>126</DHWDailyDemand>
    <InternalPartitionMass>Media</InternalPartitionMass>
    <LocationImage>ubicacion.jpg</LocationImage>
    <BuildingImage>edificio.jpg</BuildingImage>
    <AirtightnessTest>False</AirtightnessTest>
    <BlowerDoorTestQ50></BlowerDoorTestQ50>
    <BlowerDoorTestN></BlowerDoorTestN>
  </Building>

```

Figure 18: <Building> element defined as complement for the iEPB schema

```

<Building id="Building1" buildingType="MultiFamily">
  <Name>Building 1</Name>
  <StreetAddress>Avda. Ejército</StreetAddress>
  <Area>173</Area>
  <Space id="Floor1" zoneIdRef="TZ__Floor1">
    <Name>Floor1</Name>
    <Area>173.000000</Area>
  </Space>

```

Figure 19: <Building> element in the gbXML

The same happens with the data included in the “Building definition”. In this case, all this data will be defined in the <Building> element of the Spanish block

4.1.3. Envelope description

4.1.3.1. Spaces

The “zones” defined in CE3X are equivalent to the spaces in gbXML and are totally defined within the <Building> element in the gbXML as <Space> element

Datos administrativos | Datos generales | **Envolvente térmica** | Instalaciones

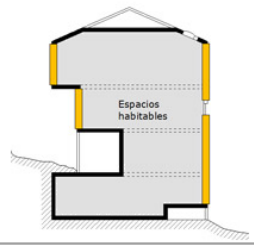
Edificio Objeto

- Floor1
- Floor2

Envolvente térmica del edificio

☐ Cubierta
☒ Muro
☐ Suelo
☐ Partición interior
☐ Hueco/Lucernario
☐ Puente térmico

☐ En contacto con el terreno
☒ De fachada
☐ Medianería



Muro de fachada

Nombre: South Facade Zona: Floor1

Dimensiones

Superficie: 25.0 m²

Longitud: 10 m

Altura: 2.5 m

Características

Orientación: Sur

Patrón de sombras: Sin patrón

Parámetros característicos del cerramiento

Propiedades térmicas Estimadas

Tipo de fachada: Una hoja

Composición del muro: 1 pie de fábrica de ladrillo

Transmitancia térmica: 0.72 W/m²K

☒ Tiene aislamiento térmico

Características del aislamiento térmico

Posición del aislamiento: Por el exterior

☒ Tipo de aislamiento: MW Espesor: 0.04 m
☐ Ra: m²K/W

Zonas

Añadir Modificar Borrar Vista clásica

Figure 20: CE3X zones definition window

Definir zonas del edificio

Definir zonas del edificio

Definición de una zona

Nombre de la zona: Floor1

Elemento raíz: Edificio Objeto

Superficie habitable asociada: 102 m²

Aceptar Cancelar

```

<Building id="Building1" buildingType="MultiFamily">
  <Name>Building 1</Name>
  <StreetAddress>Avda. Ejército</StreetAddress>
  <Area>173</Area>
  <Space id="Floor1" zoneIdRef="TZ_Floor1">
    <Name>Floor1</Name>
    <Area>102.000000</Area>
    <Volume>255.000000</Volume>
    <PeopleNumber unit="SquareMPerson">34.500000</PeopleNumber>
    <LightPowerPerArea unit="WattPerSquareMeter">4.500000</LightPowerPerArea>
    <EquipPowerPerArea unit="WattPerSquareMeter">4.500000</EquipPowerPerArea>
  </Space>
  <Space id="Floor2" zoneIdRef="TZ_Floor2">
    <Name>Floor2</Name>
    <Area>71.000000</Area>
    <Volume>177.500000</Volume>
    <PeopleNumber unit="SquareMPerson">34.500000</PeopleNumber>
    <LightPowerPerArea unit="WattPerSquareMeter">4.500000</LightPowerPerArea>
    <EquipPowerPerArea unit="WattPerSquareMeter">4.500000</EquipPowerPerArea>
  </Space>
  <BuildingStorey id="Planta_1">
    <Name>Planta 1</Name>
    <Level>0.000000</Level>
  </BuildingStorey>
</Building>

```

Figure 21: CE3X spaces definition in the gbXML inside the <Building> element

4.1.3.2. Surfaces

In CE3X tool the surfaces are defined by geometrical data + thermal characteristics of the surface.

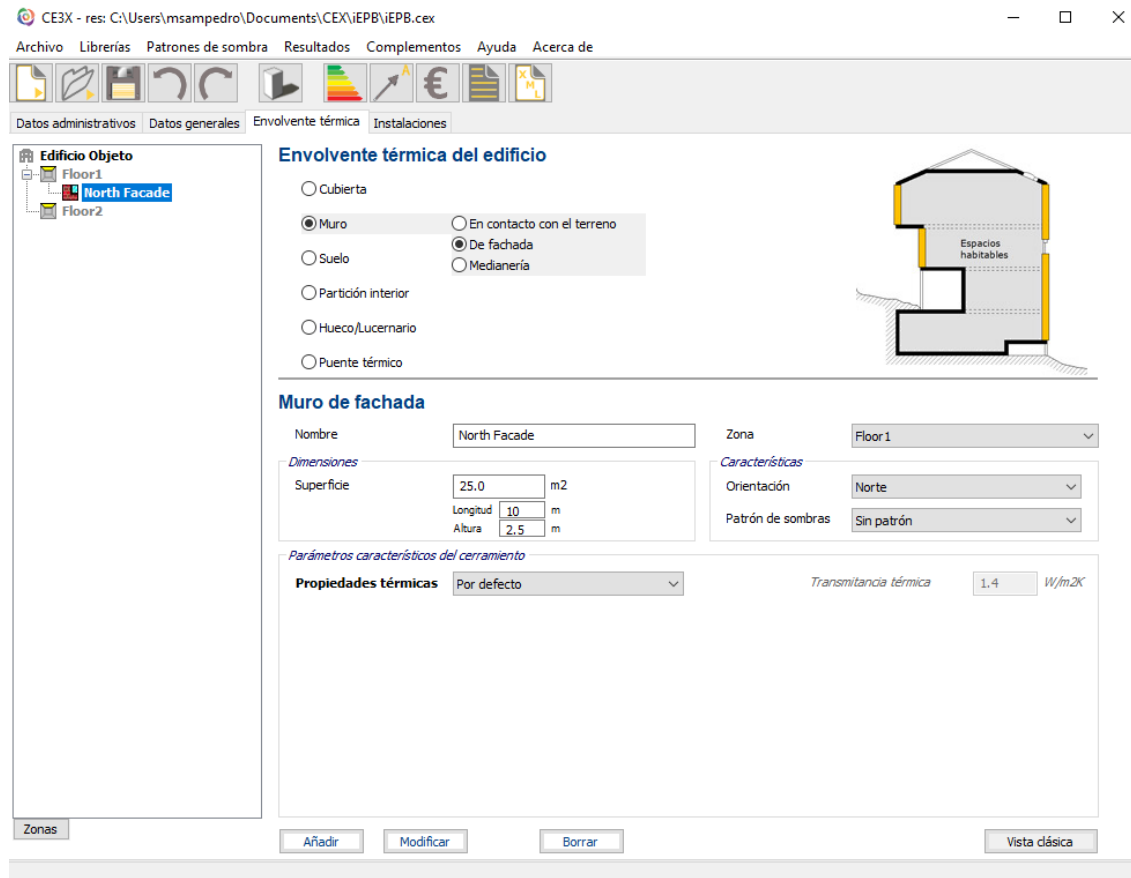


Figure 22: CE3X surface definition

The geometric data shall be defined in the gbXML block as a surface, and its thermal characteristics shall be defined by the surface attribute "constructionIdRef".

```
<Surface id="Surface_1" surfaceType="ExteriorWall" constructionIdRef="DefaultWallConstruction_1"
exposedToSun="true">
  <Name>North Facade</Name>
  <AdjacentSpaceId spaceIdRef="Floor1" />
  <RectangularGeometry>
    <Azimuth>0</Azimuth>
    <Tilt>90</Tilt>
    <Width>10</Width>
    <Height>2.5</Height>
  </RectangularGeometry>
</Surface>
```

Figure 23: North Facade geometrical data included in CE3X and defined in the gbXML schema

The way of defining the constructions shall be different in order to adapt to the different ways of defining the thermal characteristics in CE3X. For this reason, some constructions will be defined entirely in the gbXML block e.g. the features included in CE3X as "default" (where only the name of the construction and the U-value are included) or those constructions defined by layers, while the constructive solutions defined in an "estimated" way will be partially located in the gbXML block and in the Spanish block for CE3X.

```
<Construction id="DefaultWallConstruction_1">
  <Name>Default wall construction for CE3X 1</Name>
  <U-value unit="WPerSquareMeterK">1.40</U-value>
</Construction>

<Construction id="ThreeLayerConstructionExample">
  <Name>Three Layer Construction Example</Name>
  <LayerId layerIdRef="Half_Brick_Thickness_Medium_Brick_Masonry" />
  <LayerId layerIdRef="Mineral_Wool" />
  <LayerId layerIdRef="Gypsum_Plasterboard" />
</Construction>
```

Figure 24: Construction defined "by default" and "by layers" in CE3X tool and their implementation in the gbXML

The screenshot shows the 'Envoltante térmica del edificio' (Building Thermal Envelope) configuration in the CE3X tool. The interface is divided into several sections:

- Edificio Objeto:** A tree view on the left showing the building structure, including 'Floor1' and 'Floor2', and various facades like 'North Facade', 'South Facade', 'East Facade', 'West Facade 1', 'West Facade 2', and 'Flat Roof'.
- Envoltante térmica del edificio:** A central panel with radio buttons for selecting the type of construction: 'Cubierta', 'Muro' (selected), 'Suelo', 'Partición interior', 'Hueco/Lucernario', and 'Puente térmico'. For 'Muro', there are sub-options: 'En contacto con el terreno', 'De fachada' (selected), and 'Medianería'.
- Muro de fachada:** A section for facade wall details. It includes:
 - Nombre:** 'South Facade'
 - Zona:** 'Floor1'
 - Dimensiones:** 'Superficie' (25.0 m²), 'Longitud' (10 m), and 'Altura' (2.5 m).
 - Características:** 'Orientación' (Sur) and 'Patrón de sombras' (Sin patrón).
 - Parámetros característicos del cerramiento:** 'Propiedades térmicas' (Estimadas), 'Tipo de fachada' (Una hoja), and 'Composición del muro' (1 pie de fábrica de ladrillo).
 - Tiene aislamiento térmico:** A checked checkbox.
 - Características del aislamiento térmico:** 'Posición del aislamiento' (Por el exterior), 'Tipo de aislamiento' (MW), 'Espesor' (0.04 m), and 'Ra' (m²K/W).
 - Transmitancia térmica:** 0.72 W/m²K.
- Diagrama:** A small schematic on the right showing the building footprint with 'Espacios habitables' (habitable spaces) highlighted.

Figure 25: "Estimated" wall construction in CE3X tool

```
<Construction id="EstimatedWallConstruction_1">
  <Name>Estimated wall construction for CE3X 1</Name>
  <U-value unit="WPerSquareMeterK">0.72</U-value>
</Construction>
```

Figure 26: Above "Estimated" construction in CE3X tool partially located in the gbXML

```
<Construction id="EstimatedWallConstruction_1" calculationType="FachadaExteriorEstimada">
  <ConstructionType>Una hoja</ConstructionType>
  <WallComposition>1 pie de fábrica de ladrillo</WallComposition>
  <AirCavityType></AirCavityType>
  <InsulationIsPresent>True</InsulationIsPresent>
  <InsulationPosition>Por el exterior</InsulationPosition>
  <InsulationType>MW</InsulationType>
  <Thickness>0.04</Thickness>
  <ThermalResistance></ThermalResistance>
</Construction>
```

Figure 27: Above "Estimated" construction in CE3X tool partially located in the Spanish block for CE3X

4.1.3.3. Windows

The definition of the windows in the iEPB schema is less straightforward since in the gbXML schema the windows are defined between two different elements <Opening> and <WindowType>. In the case of windows in CE3X the <Opening> element of gbXML will only be used to define the dimensions of the window (width, height) and the setback. The rest of the data are defined through the gbXML <WindowType> element or through <WindowType> element defined in the

CE3X block since the values to be entered in CE3X tool are not overall values related to the opening, but partial values of its components (glass, frame, solar protections...).

Envoltente térmica del edificio

☐ Cubierta
☐ Muro
☐ Suelo
☐ Partición interior
☒ Huevo/Lucernario
☐ Puente térmico

Huevo/Lucernario

Nombre: South window 1
 Cerramiento asociado: South Facade
 Orientación: Sur

Dimensiones

Longitud	0.9	m
Altura	2.2	m
Multiplicador	3	
Superficie	5.94	m ²
Porcentaje de marco	20	%

Características

Permeabilidad del huevo: Valor conocido 27 m³/hm²
 Absortividad del marco: 0.75
☒ Dispositivo de protección solar: Dispositivo de protección solar
 Patrón de sombras: Sin patrón
☐ Doble ventana

Parámetros característicos del huevo

Propiedades térmicas Estimadas

Tipo de vidrio	Doble	U vidrio	3.3	W/m ² K
Tipo de marco	Metálico sin RPT	g vidrio	0.75	
		U marco	5.7	W/m ² K

Figure 25: Window by “estimated” calculation introduced in CE3X tool

```
<Surface id="Surface_2" surfaceType="ExteriorWall" constructionIdRef="EstimatedWallConstruction_1" exposedToSun="true">
  <Name>South Facade</Name>
  <AdjacentSpaceId spaceIdRef="Floor1" />
  <RectangularGeometry>
    <Azimuth>180</Azimuth>
    <Tilt>90</Tilt>
    <Width>10</Width>
    <Height>2.5</Height>
  </RectangularGeometry>
  <Opening id="Window_1" windowTypeIdRef="EstimatedWindow1" openingType="OperableWindow">
    <Name>South window 1</Name>
    <RectangularGeometry>
      <Azimuth>180</Azimuth>
      <Tilt>90</Tilt>
      <Width>0.9</Width>
      <Height>2.2</Height>
    </RectangularGeometry>
    <Setback>0.2</Setback>
  </Opening>
</Surface>

<WindowType id="EstimatedWindow1">
  <Name>DefaultDefinitionWindow 1</Name>
  <U-value unit="WPerSquareMeterK">3.3</U-value>
  <SolarHeatGainCoeff unit="Fraction">0.75</SolarHeatGainCoeff>
  <Blind id="Overhang_1"/>
  <Blind id="HorizontalBlinds_1"/>
  <Frame id="Frame_1" type="Aluminum">
    <Name>Metálico Sin RPT</Name>
  </Frame>
</WindowType>
```

Figure 26: Window definition in the gbXML as <Opening> and <WindowType>


```

<WindowType id="EstimatedWindow1">
  <GlazeType>Doble</GlazeType>
  <FrameType>Metálico sin RPT</FrameType>
</WindowType>

<Opening id="Window_1">
  <Multiplier>3</Multiplier>
  <Area unit="SquareMeters">14.4</Area>
  <FrameRate unit="Percent">20</FrameRate>
  <WindowAirPermeabilityDefinition>Valor conocido</WindowAirPermeabilityDefinition>
  <WindowAirPermeability>27</WindowAirPermeability>
</Opening>

```

Figure 27: Complementary window definition in the the Spanish CE3X block

4.1.3.4. Thermal Bridges

The gbXML schema does not include any element for the definition of linear and point thermal bridges of buildings. In the Spanish CE3X program it is necessary to introduce the linear thermal transmittances of the most common constructive encounters of the buildings, for this reason these elements will be defined in its block or, in case of sharing data with the CERMA program, these data will be defined partially or totally in the certification block or partially or totally in the CE3X program block.

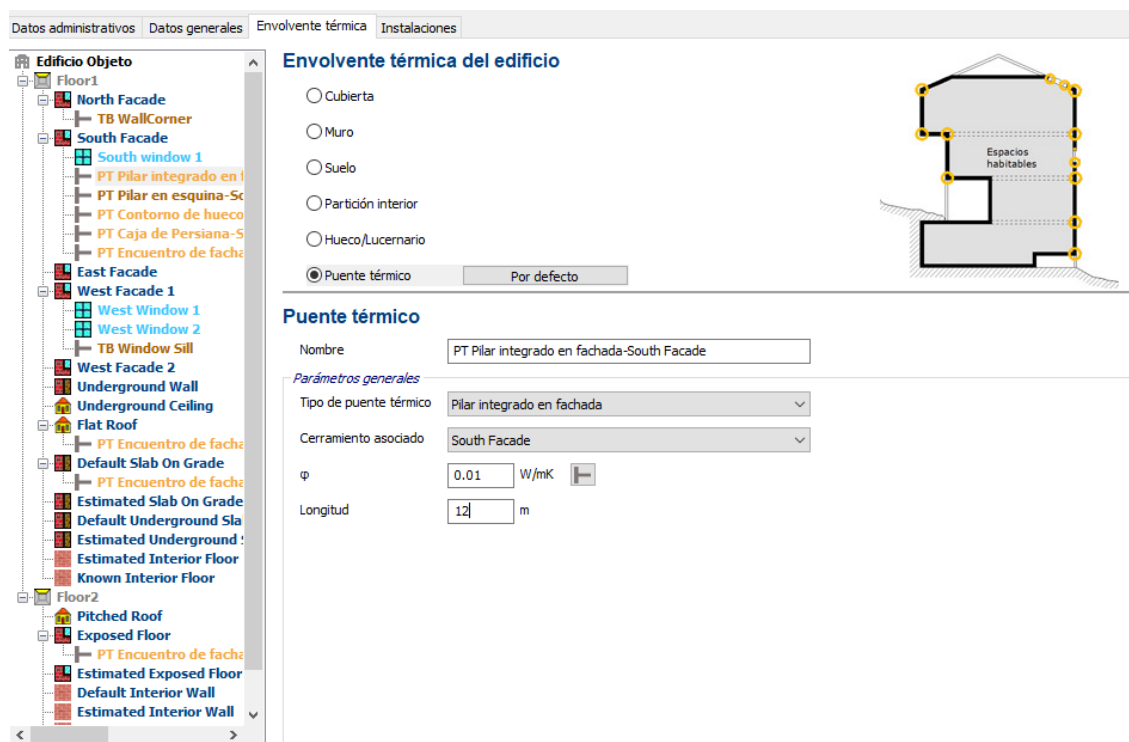


Figure 28: Thermal bridge “PT Pilar integrado en fachada-South Facade” defined in CE3X tool

```

<ThermalBridge id="ThermalBridge_1" type="FacadeIntegratedPillar">
  <Name>PT Pilar integrado en fachada-South Facade</Name>
  <FamilyName>Pilar integrado en fachada</FamilyName>
  <AdjacentSurfaceId surfaceIdRef="Surface_2"></AdjacentSurfaceId>
  <Psi unit="WPerMeterK">0.01</Psi>
  <Length unit="Meters">12</Length>
</ThermalBridge>

<ThermalBridge id="ThermalBridge_2" type="CornerPillar">
  <Name>PT Pilar en esquina-South Facade</Name>
  <FamilyName>Pilar en esquina</FamilyName>
  <AdjacentSurfaceId surfaceIdRef="Surface_2"></AdjacentSurfaceId>
  <Psi unit="WPerMeterK">0.17</Psi>
  <Length unit="Meters">6</Length>
</ThermalBridge>

```

Figure 29: Thermal bridge “PT Pilar integrado en fachada-South Facade” definition in the CE3X block

4.1.4. Building facilities

The procedure for the definition of the building facilities will be the same as for the building envelope. All possible data will be defined in the gbXML and the rest will go to the assessment or tool block.

Instalaciones del edificio

☐ Equipo de ACS
 ☐ Contribuciones energéticas
☐ Equipo de sólo calefacción
 ☐ Equipos de iluminación
☐ Equipo de sólo refrigeración
 ☐ Equipos de aire primario
☐ Equipo de calefacción y refrigeración
☒ Equipo mixto de calefacción y ACS
☐ Equipo mixto de calefacción, refrigeración y ACS

Equipo mixto de calefacción y ACS

Nombre: Calefacción y ACS (40%)/Gas Zona: Edificio Objeto

Características

Tipo de generador: Caldera Estándar
Tipo de combustible: Gas Natural

Demanda cubierta

	ACS	Calefacción
Superficie (m2)	173.0	173.0
Porcentaje (%)	100	100

Rendimiento medio estacional

Rendimiento estacional: Estimado según Instalación Rendimiento medio estacional (ACS y Calefacción): 56.8 %

Potencia nominal: 24.0 kW
Carga media real ómb: 0.2 ?
Rendimiento de combustión: 85 %

Aislamiento de la caldera: Antigua con mal aislamiento

☐ Con Acumulación

Zonas: Añadir Modificar Borrar Vista clásica

Figure 30: “DHW and heating equipment with natural gas boiler” definition in CE3X

```

<gbXML xmlns="http://www.gbxml.org/schema" xmlns:xhtml="http://www.w3.org/1999/xhtml"
  <HydronicLoop id="HydronicLoop" loopType="HotWater" fluidType="Water">
    <HydronicLoopEquipment id="HydronicLoopEquipment_1" equipmentType="Boiler">
      <Power unit="Kilowatt" powerType="NaturalGas" useType="Heating">24</Power>
      <Efficiency efficiencyType="HSPF">0.568</Efficiency>
    </HydronicLoopEquipment>
  </HydronicLoop>

```

Figure 31: “DHW and heating equipment with natural gas boiler” definition in the gbXML

```

<iEPBxml xmlns="http://www.efinovatic.es/ce3x" xmlns:xhtml="http://www.w3.org/1999/xhtml" xmlns:xsi="http://www.w3.org
<Spain>
  <EPC>
    <CE3X order = "0">
      <HydronicLoopEquipment id="HydronicLoopEquipment_1" equipmentType= "Equipo mixto de calefacción y ACS"
        calculationType = "Estimado según instalación">
          <Zone></Zone>
          <DHWCoverage unit="SquareM">173</DHWCoverage>
          <HeatingCoverage unit="SquareM">173</HeatingCoverage>
          <GeneratorType>Caldera Estándar</GeneratorType>
          <CombustionLoadFactor>0.2</CombustionLoadFactor>
          <CombustionEfficiency units="Percent">85</CombustionEfficiency>
          <BoilerInsulationLevel>Antigua con mal aislamiento</BoilerInsulationLevel>
          <DHWTankPresent>False</DHWTankPresent>
          <DHWUA></DHWUA>
          <Multiplier></Multiplier>
          <UACalculationType></UACalculationType>
          <HighTemperature unit="C"></HighTemperature>
          <LowerTemperature unit="C"></LowerTemperature>
          <TankVolume unit="CubicMeters"></TankVolume>
          <InsulationType></InsulationType>
          <Thickness unit="Meters"></Thickness>
        </HydronicLoopEquipment>

```

Figure 32: “DHW and heating equipment with natural gas boiler” definition in the CE3X block

4.2. Austrian software Ecotech to iEPB

The following describes an example of the implementation of Austrian ECOTECH software into the iEPB format. This implementation includes interesting aspects that should be taken into account for harmonisation at EU level.

4.2.1. Import/Export:

Since surfaces typically represent the largest data group within an energy label, they are used as an example to illustrate the import/export process.

Import a .iEPB file:

If the node */iEPBxml/Austria/EPC/ecotech* does not exist in the iEPB.xml file, the importer attempts to extract the relevant data from the gbXML file and creates the */iEPBxml/Austria/EPC/ecotech* node.

In the gbXML format, areas, volumes, and coordinates are defined based on internal and centroid measurements. However, Austrian regulations require that all building measurements be taken from the exterior. Therefore, when importing a model from gbXML, all measurements are interpreted as external values, even though the original gbXML data represents internal dimensions.

Export a .iEPB file:

When exporting an iEPB file, a gbXML file is generated that includes all data available in an Austrian EPC calculation and compatible with the gbXML schema.

For example, all surfaces are exported with their corresponding area. However, since ecotech does not contain 3D information, the coordinates (points) defining these areas are not exported. The measures in this case are exported as external areas and volumes.

On the other hand, information related to shading or detailed ground (earth) heat losses is not represented in gbXML and therefore is not included in the exported gbXML file.

Baukörper [Saniert mit Garage 25.02.2025 15:22:02 - Variante 1 21.10.2025 13:34:44]

Stammdaten **Außenwand** Innenwand Fußboden Decke Dach Volumen Wärmebrücke

Bezeichnung	Brutto-Fl.	Netto-Fl.
AW-EG Nord	41,14 m²	33,39 m²
AW-EG Ost	11,70 m²	11,70 m²
AW-EG Süd	41,14 m²	33,82 m²
AW-EG West	26,40 m²	25,67 m²
AW-OG Nord	41,94 m²	36,75 m²
AW-OG Ost	26,92 m²	26,92 m²
AW-OG Süd	41,94 m²	36,01 m²
AW-OG West	26,92 m²	24,92 m²

Erläuterungen zum Erfassen von Außenwänden

Bezeichnung: AW-EG Nord
Ausrichtung / Typ erdanliegend: Nord
Verwendeter Bauteil: AW 0,51m U=0,15
U-Wert: 0,15 W/m²K
Dicke: 0,512m
Anzahl: 1
Breite [m]: 13,4
Höhe [m]: 3,07
Zustand: warm / außen
Eingaben für Detaillierte Erdverluste
Eingaben für Verschattungsfaktoren

Anz	Bezeichnung	Einzel-Fl.	Gesamt-Fl.
1	AF 1,33/1,00m U=0,79	-1,33 m²	-1,33 m²
1	AF 2,00/2,16m U=0,79	-4,32 m²	-4,32 m²
1	AF 0,97/2,16m U=0,79	-2,10 m²	-2,10 m²

Hinzufügen Bearbeiten Löschen

Korrektur Fläche: 7,75 m²

Neue Fläche Fläche Löschen

BGF=230,48 m² Beheiztes Volumen=708,73 m³ A/V=0,71 m²/m³

Figure 33: Representation of a surface in ecotech

```

<Surface constructionIdRef="cnstr-2" exposedIosun="true" id="su-3" surfacetype="ExteriorWall">
  <Name>Wand-002</Name>
  <RectangularGeometry>
    <Azimuth>90.00000000</Azimuth>
    <Tilt>90.00000000</Tilt>
    <Width>4.43500001</Width>
    <Height>2.58000003</Height>
    <CartesianPoint>
      <Coordinate>10.57000000</Coordinate>
      <Coordinate>0.86499999</Coordinate>
      <Coordinate>0.00000000</Coordinate>
    </CartesianPoint>
  </RectangularGeometry>
  <PlanarGeometry>
    <PolyLoop>
      <CartesianPoint>
        <Coordinate>10.57000000</Coordinate>
        <Coordinate>5.30000000</Coordinate>
        <Coordinate>2.58000003</Coordinate>
      </CartesianPoint>
      <CartesianPoint>
        <Coordinate>10.57000000</Coordinate>
        <Coordinate>0.86499999</Coordinate>
        <Coordinate>2.58000003</Coordinate>
      </CartesianPoint>
      <CartesianPoint>
        <Coordinate>10.57000000</Coordinate>
        <Coordinate>0.86499999</Coordinate>
        <Coordinate>0.00000000</Coordinate>
      </CartesianPoint>
      <CartesianPoint>
        <Coordinate>10.57000000</Coordinate>
        <Coordinate>5.30000000</Coordinate>
        <Coordinate>0.00000000</Coordinate>
      </CartesianPoint>
    </PolyLoop>
  </PlanarGeometry>
  <AdjacentSpaceId spaceIdRef="sp-1-Zone"/>
</Surface>

```

Figure 34: gbXML data exported from CAD

```

<Surface id="FL-ECT-20100412XXXX-ATCL0070-2025-09-24-12-29-40-4498054-166" surfacetype="ExteriorWall" constructionIdRef="BT-ECT-20100412XXXX-ATCL0004-2011-05-10-21-43-58-7823880-875">
  <Name>AW-EG Nord</Name>
  <AdjacentSpaceId spaceIdRef="ID_138"/>
  <AdjacentSpaceId spaceIdRef="ECTGEOMETRYUNIT_AUSSEN_UNIQUEID"/>
  <RectangularGeometry>
    <Azimuth>0</Azimuth>
    <Tilt>90</Tilt>
    <Width>41.1379967</Width>
    <Height>1</Height>
  </RectangularGeometry>
  <Opening id="ID_2bb142d2-b188-4086-9bbd-94873148f8ae" openingType="OperableWindow" windowTypeRef="BT-ECT-20100412XXXX-ATCL0004-2011-05-10-21-43-45-7822508-799">
    <Name>AF 1,33/1,00m U=0,79</Name>
    <U-value unit="WPerSquareMeterK">0.79000002145767212</U-value>
    <RectangularGeometry>
      <Azimuth>0</Azimuth>
      <Tilt>90</Tilt>
      <Width>1.33</Width>
      <Height>1</Height>
    </RectangularGeometry>
  </Opening>
</Surface>

```

Figure 35:gbXML data exported from ecotech (no coordinates)

```

<Item ID="FL-ECT-20100412XXXX-ATCL0070-2025-09-24-12-29-40-4498054-166">
  <ECTConstructionID>BT-ECT-20100412XXXX-ATCL0004-2011-05-10-21-43-58-7823880-875</ECTConstructionID>
  <ECTGeometryUnit1ID>ID_138</ECTGeometryUnit1ID>
  <ECTGeometryUnit2ID>ECTGEOMETRYUNIT_AUSSEN_UNIQUEID</ECTGeometryUnit2ID>
  <Name>AW-EG Nord</Name>
  <BruttoArea>
    <Item ID="">
      <Name>AW-EG Nord</Name>
      <Type>Direkt</Type>
      <Sign>1</Sign>
      <Parameters>
        <Area>41.1379967</Area>
      </Parameters>
    </Item>
  </BruttoArea>
  <PerimeterToGround>0</PerimeterToGround>
  <Orientation>0</Orientation>
  <Pitch>90</Pitch>
  <BoundingDim1>13.4</BoundingDim1>
  <BoundingDim2>3.07</BoundingDim2>
  <GenericParameters>

```

Figure 36:iEPB data of the same surface (additional field "PerimeterToGround")

4.2.2. Project data

Projekt - Erfassung

Standard | Allgemeine Berechnungseinstellungen | Energieausweis lt. OIB RL 6 | Einreichung | OIB Index | Personen | Daten (alt)

Projektbezeichnung: Leibnizstr. 20

Baujahr: 1960

Letzte Veränderung:

PLZ: 4061 Ort: Pasching

Strasse: Leibnizstr

Hausnummer: 20 Stiege, Haus, Parzelle, etc.: Tür, Top:

Gemeinde: Pasching

Grundstücksnr.: 1784/34 Einlagezahl: 971

Katastralgemeinde: Pasching KG-Nr.: 45308

Klimaregion: N - Region Nord - außerhalb von Föhngebieten

Seehöhe: 275 [m] Bereich für Seehöhe: 275 - 361 m

Normaußentemperatur: -14,1 [°C]

Normaußentemp. NAT-13: 22,5 [°C]

Daten für Energieberatung (Advice-Datei) importieren

Standortfestlegung

Figure 37:Representation of project data in ecotech

```
<ProjectData xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <Name>Leibnizstr. 20</Name>
  <BuildingPartOfBuilding>
    <Street>Leibnizstr</Street>
    <Plz>4061</Plz>
    <Site>Pasching</Site>
    <PropertyNumber>1784/34</PropertyNumber>
    <Baujahr>1960</Baujahr>
    <RecentChange/>
    <CadastralCommunityName>Pasching</CadastralCommunityName>
    <CadastralCommunityNumber>45388</CadastralCommunityNumber>
    <HeightAboveSeaLevel>275</HeightAboveSeaLevel>
    <GWRNumber/>
    <IssueDate>04.06.2013</IssueDate>
    <ValidityDate>04.06.2023</ValidityDate>
    <ContentCreator>BuildDesk Österreich GmbH Alfred Bruckmoser</ContentCreator>
    <theta_e>-14.1</theta_e>
    <theta_m>e</theta_m>
  </BuildingPartOfBuilding>
</ProjectData>
```

Figure 38:iEPB data of the project

XPATH: `/iEPBxml/Austria/EPC/ecotech/ECTEPBDProject/ProjectData`

The screenshot shows a software window titled "Projekt - Erfassung" with a tabbed interface. The "Daten (alt)" tab is active. It contains two main sections: "Eigentümer" (Owner) and "ErstellerIn" (Creator). The "Eigentümer" section displays the following information: Herr Markus Maier, MeMyselfAndI, Dorfstr 10, A-4020 Linz, Telefon: +43 12345678, E-Mail: Markus.Maier@MeMyselfAndI.com. Below this is a button labeled "Eigentümer bearbeiten". The "ErstellerIn" section displays: Alfred Bruckmoser, BuildDesk Österreich GmbH, Kapuzinerstr. 84e, A-4020 Linz, Telefon: 0732 77 43 24. Below this is a button labeled "Ersteller bearbeiten". At the bottom, there are input fields for "Gebäudezone/Gebäude(-teil)", "GWR-Zahl", and "Geschäftszahl", and dropdown menus for "Ausstellungsdatum" (set to 04.06.2013) and "Gültigkeitsdatum" (set to 04.06.2023). At the very bottom are "Abbrechen" and "OK" buttons.

Figure 39:Person data of the owner in ecotech

```
<PersonOwner>
  <Salutation>Herr</Salutation>
  <Title/>
  <FirstName>Markus</FirstName>
  <Surname>Maier</Surname>
  <Company>MeMyselfAndI</Company>
  <Telephone>+43 12345678</Telephone>
  <Fax/>
  <Email>Markus.Maier@MeMyselfAndI.com</Email>
  <State>A</State>
  <PostalCode>4020</PostalCode>
  <Site>Linz</Site>
  <Street>Dorfstr</Street>
  <HouseNumber>10</HouseNumber>
</PersonOwner>
```

Figure 40:iEPB data of the owner

XPATH: `/iEPBxml/Austria/EPC/ecotech/ECTEPBDProject/ProjectData/PersonOwner`

4.2.3. Constructions

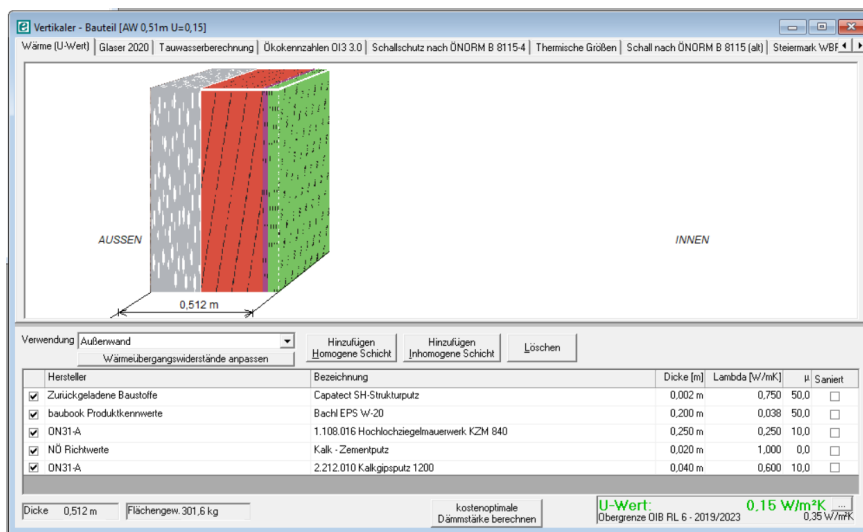


Figure 41: Construction in Ecotech

In Austria, ecological performance calculations are required to obtain a building permit and financial subsidies. These calculations must include additional indicators: Global Warming Potential (GWP), Primary Energy Consumption, and Acidification Potential. These indicators are determined based on predefined construction materials.

To enable this mapping in gbXML files, users must link the constructions defined in the gbXML file to the corresponding pre-configured constructions in ecotech. Only after this assignment can the ecological indicators be correctly applied and evaluated.

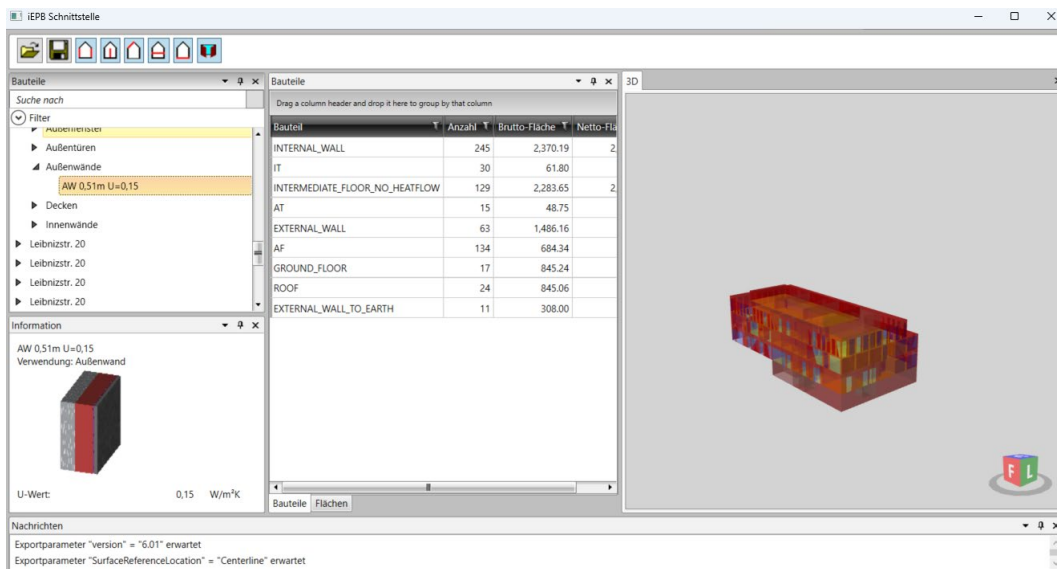


Figure 42: Assigning Construction to a gbXML file.


```

<ECTConstructions>
  <Item ConstructionGroupID="" ID="BT-ECT-20101006XXXA-DESKTOP-1E1VDM-2025-07-29-09-27-26-3404658-722">
    <ConstructionType>EXTERNAL_WALL</ConstructionType>
    <Name>AW H3 Sanierung Außenwand</Name>
    <Info>
      <DateChanged>2025-07-29T09:59:05</DateChanged>
      <DateCreated>2025-07-29T09:27:26</DateCreated>
    </Info>
    <Addons>
      <Layers>
        <Item ID="">
          <Name>RÜFIX Sisi-Putz VITAL</Name>
          <LayerType>homogeneous</LayerType>
          <Thickness>0.005</Thickness>
          <Materials>
            <Item MaterialID="2142685311" BinderMaterialID="">
              <Polygon>
                <Item><X>0</X><Y>0</Y></Item>
                <Item><X>1</X><Y>0</Y></Item>
                <Item><X>1</X><Y>1</Y></Item>
                <Item><X>0</X><Y>1</Y></Item>
              </Polygon>
              <BinderCount>0</BinderCount>
              <Airgap>noAirgap</Airgap>
              <YearOfConstruction>1965</YearOfConstruction>
            </Item>
          </Materials>
          <IsCutToFullLayer>false</IsCutToFullLayer>
          <CalcThermic>FLEX_CHECKBOX_STATE_CHECKED</CalcThermic>
          <Epsilon1>0</Epsilon1>
          <Epsilon2>0</Epsilon2>
          <Epsilon1Type>Normal_high_emissivity</Epsilon1Type>
          <Epsilon2Type>Normal_high_emissivity</Epsilon2Type>
          <deltaT>0</deltaT>
          <deltaTType>default</deltaTType>
          <MeanTemperatureType>M10</MeanTemperatureType>
          <CalcI3>FLEX_CHECKBOX_STATE_CHECKED</CalcI3>
          <Renovated>false</Renovated>
        </Item>
      </Layers>
    </Item>
  </Item>

```

Figure 43: Construction in iEPB

XPATH: /iEPBxml/Austria/EPC/ecotech/ECTEPBDProject/ECTClasslib/ECTConstructions/Item[1]

4.2.4. HVAC-Data

In the gbXML format, the HVAC structure does not align with the Austrian system structure. Many HVAC system types commonly used in Austria (e.g. heat pumps) are not represented in gbXML (for example, heat pumps are only available in the AirLoop section). As a result, the gbXML file does not contain the necessary information for these systems.

Raumheizung - Berechnung nach ÖNORM H 5056

Raumheizung - Wärmebereitstellung

- ☒ Zentrale Wärmebereitstellung ☐ Dezentrale Wärmebereitstellung
- Art Bereitstellung: Bivalente Wärmepumpe (Alternativbetrieb)
- Produkt: Vaillant Austria GmbH flexo THERM VWF 117/4 LW
- Quell-/Heizungsmedium: Außenluft / Wasser (A7/W35)
- Nennleistung [kW]: 9,0 (Default) / 11,4 (Freie Eingabe)
- COP am Prüfpunkt: 3,96 (Default) / 4,6 (Freie Eingabe)
- Bivalenzpunkt [°C]: -4,1 (Default) / -10 (Freie Eingabe)
- ☒ E-Heizung als 2. System

Raumheizung - Wärmespeicherung

- Speicher: Pufferspeicher Solar (ohne WW; 55 °C)
- Produkt:
- Anschlussteile: Basisanschlüsse, E-Patrone, Heizregister für Solaranlage
- ☐ Speicher im beheizten Bereich

Raumheizung - Wärmeverteilung

	Lage	Dämmung Leitungen	Dämmung Armaturen	Länge [m]
Verteilleitungen	50% beheizt	2/1 Durchmesser	Armaturen gedämmt	16,35
Steigleitungen	Unbeheizt	2/3 Durchmesser	Armaturen gedämmt	18,44
Anbindelösungen	100% beheizt	1/3 Durchmesser	Armaturen gedämmt	64,53

Verteilkreisregelung: Gleitende Betriebsweise

☐ Zweileitersystem

Raumheizung - Wärmeabgabe

- Art der Regelung: Raumthermostat-Zonenregelung mit Zeitsteuerung
- Abgabesystem: Flächenheizung (35/28 °C)
- Ermittlung: Individuelle Verbrauchsermittlung und Heizkostenabrechnung (Fixwert)

Figure 44: Settings for room heating in ecotech

```

<Heating>
  <optCentral>true</optCentral>
  <txtBGF>
    <Wert>0</Wert>
    <FreieEingabe>false</FreieEingabe>
    <Default>230.480011</Default>
  </txtBGF>
  <HeatEmission>
    <cmdControl>RoomThermostat</cmdControl>
    <cmbEmissionSystem>PanelHeating3528</cmbEmissionSystem>
    <cmbConsumptionDetermination>Individual</cmbConsumptionDetermination>
  </HeatEmission>
  <HeatDistribution>
    <freieEingabe>false</freieEingabe>
    <cmbDistributionPipesPosition> 50heated</cmbDistributionPipesPosition>
    <cmbRisingPipesPosition> unheated</cmbRisingPipesPosition>
    <cmbConnectionPipesPosition> 100heated</cmbConnectionPipesPosition>
    <cmbDistributionPipesInsulation> 21insulated</cmbDistributionPipesInsulation>
    <cmbRisingPipesInsulation> 23insulated</cmbRisingPipesInsulation>
    <cmbConnectionPipesInsulation> 13insulated</cmbConnectionPipesInsulation>
    <cmbDistributionPipesMountings>Insulated</cmbDistributionPipesMountings>
    <cmbRisingPipesMountings>Insulated</cmbRisingPipesMountings>
    <cmbConnectionPipesMountings>Insulated</cmbConnectionPipesMountings>
    <cmbDistributionPipesLength>
      <Wert>0</Wert>
      <FreieEingabe>false</FreieEingabe>
      <Default>16.3504314</Default>
    </cmbDistributionPipesLength>
    <cmbRisingPipesLength>
      <Wert>0</Wert>
      <FreieEingabe>false</FreieEingabe>
      <Default>18.4384</Default>
    </cmbRisingPipesLength>
  </HeatDistribution>
</Heating>

```

Figure 45:iEPB data for room heating

XPATH: /iEPBxml/Austria/EPC/ecotech/ECTEPBDProject/GEEInput/Heating

4.3. SRI2MARKET web tool to iEPB

The SRI2MARKET tool is an online application developed by the SRI2MARKET project co-funded by the European Union. It is compliant with the legal framework of the Smart Readiness Indicator (SRI), a common EU scheme for rating the smart readiness of buildings. It implements the national technical frameworks decided by 6 EU countries represented in the SRI2MARKET project.

Although the iEPB project did not contemplate the implementation of an SRI assessment tool in the iEPB schema, the development and fulfillment of its various tasks have generated the need to implement an SRI assessment tool in the format. We assume that the incorporation of an SRI assessment tool was not considered in the schema due to its absence in the market at the time the proposal was drafted.

Taking advantage of the fact that two of the iEPB project partners (EFINOVATIC and CENER) were responsible for developing the SRI2MARKET tool in the SRI2MARKET project, the opportunity arose to include the tool in the iEPB schema.

Therefore, the SRI2MARKET tool is one of the tools that is already part of the iEPB schema. The following has been implemented in this web tool:

4.3.1. Import/Export:

The import of an iEPB file into SRI2MARKET web tool, is done from the tab for creating a “New project”. At the bottom of the tab, there is a box where you can drag and drop an iEPB file to import its data.

The screenshot displays the 'New project' tab in the SRI2MARKET web tool. The form includes fields for 'Description of building', 'Address', and 'Energy class'. Below these is a 'Methology Selection' section with checkboxes for various building systems: Heating, DWH, Cooling, Ventilation, Lighting, Dynamic Building Envelope, Electricity, Electric Vehicle Charging, and Monitoring And Control. At the bottom, a red box highlights the 'Import iEPB File' section, which contains a large text area with the instruction 'Drag & drop your iEPB file here, or click to browse' and a 'Browse' button. The footer of the page includes logos for 'Designed by', 'CONTACT INFO', 'DISCLAIMER', and the European Union flag.

Figure 46: SRI2MARKET *New project* tab

The export of data from SRI2MARKET to an iEPB file. SRI assessment files can be exported to iEPB format from the ‘Project List’ tab of the SRI2MARKET web tool using the ‘Download’ button.

SRI2MARKET

Welcome msampedro Sign off Language: English (en)

Home New project Project List Basic Benchmarking Advanced Benchmarking Quick Exploration FAQs

Project List

Name	Catalogue	Country	Modify	Delete	Duplicate	ISO52120	ISO52120	Download
iEPB example new	Default Method B	Spain	Modify	Delete	Duplicate	↑ A	↑ B	Download

1 / 1

Designed by: CENER, efinovatic

CONTACT INFO: <https://learning.sri2market.eu/>, Email: contact@sri2market.eu

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Figure 47: SRI2MARKET web Tool *Project List* tab

4.3.2. SRI assessment data

As the SRI assessment is a qualitative assessment that does not have geometric data on the building or technical specifications for materials or construction, it has little data shared with the gbXML format, and almost all of the stored data is found in the iEPB.xml file. However, the criteria still apply.

The data recorded in the gbXML.xml and iEPB.xml files is shown below.

```

1 <gbXML xmlns="http://www.gbxml.org/schema" xmlns:xhtml="http://www.w3.org/1999/xhtml" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" x
2 <Campus id="Facility">
3   <Building id="12" buildingType="Unknown">
4     <Name>CENER headquarters</Name>
5     <Area>5253.0</Area>
6     <Description>Bioclimatic office building</Description>
7   </Building>
8 </Campus>
9 <DocumentHistory>
10  <CreatedBy programId="SRI2Market" date="2024-04-24T14:28:10" personId="2" />
11  <ProgramInfo id="Sri2Market">
12    <Product>Sri2Market</Product>
13    <Version>0.0.0</Version>
14    <Platform>Unknown</Platform>
15  </ProgramInfo>
16  <PersonInfo id="2">
17    <FirstName>Maria</FirstName>
18    <LastName>Fernandez</LastName>
19  </PersonInfo>
20 </DocumentHistory>
21 </gbXML>

```

Figure 48: SRI data stored in the <Campus> and <DocumentHistory> elements of the gbXML.xml file.

```

1 <iEPB xmlns="http://www.efinovatic.es/sri" xmlns:xhtml="http://www.w3.org/1999/xhtml" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instanc
2 <Spain>
3   <Building id="12">
4     <YearOfConstruction>2000.0</YearOfConstruction>
5   </Building>
6   <SRI>
7     <Results>
8       <TotalSriScore>32.51</TotalSriScore>
9       <ScoreKF1>48.561499722209625</ScoreKF1>
10      <ScoreKF2>37.752416301291966</ScoreKF2>
11      <ScoreKF3>11.22023160684932</ScoreKF3>
12    </Results>
13    <SRI2MARKET version="0.1">
14      <Projects>
15        <Project id="12">
16          <AssessmentPurpose>Current building state</AssessmentPurpose>
17          <Organisation>CENER</Organisation>
18          <EmailAddress>contact@sri2market.eu</EmailAddress>
19          <TelephoneNumber></TelephoneNumber>
20          <NetFloorAreaBuilding>5253.0</NetFloorAreaBuilding>
21          <BuildingState>Original</BuildingState>
22          <EffectiveRatedOutputHVACSystem>4gt;= 290 kW</EffectiveRatedOutputHVACSystem>
23          <DescriptionBuilding>Bioclimatic office building</DescriptionBuilding>

```

Figure 49: SRI data stored in the iEPB.xml file.

5. Conclusions

The development of the iEPB schema represents a significant step towards harmonizing building energy performance assessments in Europe. By establishing a common data format, it facilitates interoperability among energy certification tools, enabling more efficient integration of various EPB instruments, such as energy certificates, smart readiness indicators, and renovation passports.

The modular structure of the schema, based on XML files and specific folders for complementary information, allows for flexible and scalable data management. This architecture makes it possible to incorporate future tools and methodologies without having to redesign the system from scratch.

The approach adopted for defining elements and attributes, as well as the differentiation among common and country-specific blocks, assessment blocks, and software blocks, ensures high quality in the organization of data. This standardization facilitates both human reading and automatic processing, which is key to its adoption by developers and public administrations.

The practical implementation of the schema, exemplified by the CE3X, Ecotech, and SRI2MARKET software, demonstrates its real applicability and its ability to adapt to the specific needs of each tool. The ability to store multiple assessments for the same building, including improvement proposals, reinforces its usefulness as an energy monitoring tool.

In short, the iEPB schema lays the foundations for a digital transformation in the field of building energy certification, promoting efficiency, transparency, and collaboration among stakeholders in the sector. Its future publication as an open repository will allow for its scope to be expanded and enable its adoption as a European standard for EPB data management.



Release Notes

Version v1.6

- The naming convention for files and folders contained in the iEPB format has been included.
- The compression algorithm to be used for the generation of the iEPB format is included.
- A release section is added to the guide to record changes in each version.

Version v1.7

- An executive summary has been added to the document.
- Conclusions have been added to the document.
- The example of implementation of the Austrian software Ecotech has been added to the examples.
- The example of implementation of the web tool SRI2MARKET has been added to the examples.

